

David Xia

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RESEARCH INTERESTS

Theoretical ML, reinforcement learning, multi-armed bandits, optimization, high-dimensional probability

EDUCATION

• University of Illinois Urbana-Champaign

Aug 2023 - May 2027

Bachelor of Science in Computer Science

Champaign, Illinois

Bachelor of Science in Liberal Arts & Sciences Major in Mathematics, Data Optimization Concentration

Bachelor of Science in Liberal Arts & Sciences Major in Statistics

◦ GPA: 3.98/4.00

PAPERS

SUB=SUBMITTED, ACC=ACCEPTED

2025 [SUB] RSK linear operators and the Vershik-Kerov-Logan-Shepp curve [arXiv]

Duy Phan, David Xia

Submitted to *Electronic Journal of Combinatorics*

2024 [ACC] EveGuard: Defeating Vibration-based Side-Channel Eavesdropping with Audio Adversarial Perturbations [arXiv]

Jung-Woo Chang, Ke Sun, David Xia, Xinyu Zhang, Farinaz Koushanfar

IEEE Symposium on Security and Privacy, 2025

WORKSHOP PAPERS & MANUSCRIPTS

SUB=SUBMITTED, ACC=ACCEPTED

2026 [SUB] Predictability in Major League Sports: Betting Odds versus Mathematical Models

David Xia, AJ Hildebrand

Submitted to the *Mathematics and Sports (MAS) Journal*

2026 [ACC] Revitalizing Local Democracy: A Human-Centered Audit of LLMs in City Council Journalism

David Xia, Chris Maury

To appear in *Human-centered Evaluation and Auditing of Language Models (HEAL) Workshop at CHI 2026*

RESEARCH EXPERIENCE

• Illinois Combinatorics Lab for Undergraduate Experience (ICLUE), Research Mentee Aug 2024 - present

- Advised by Professor Alexander Yong.
- Proved a new asymptotic result that the probability that the Schensted insertion algorithm for a uniformly random permutation in S_n exhibits a special "bumping" interaction converges to 1 as n approaches infinity.
- Derived the result using rigorous probability bounds on the RSK linear operator by leveraging the Vershik-Kerov-Logan-Shepp limit shape curve as the central analytic tool.
- Developed proofs requiring techniques spanning real analysis (uniform convergence, limit arguments), probability theory (concentration inequalities, convergence in probability), and asymptotic analysis (bounding combinatorial quantities in the large- n regime).
- Paper in submission to the [Electronic Journal of Combinatorics](#).

• University of California San Diego, Research Intern

Jun 2023 - Dec 2024

- Advised by Professor Xinyu Zhang.
- Developed EveGuard, a software-based defense framework to protect voice privacy from vibrometry-based side channels using adversarial audio by using a perturbation generator model (PGM) to effectively suppress sensor-based eavesdropping while preserving high audio quality.

- Implemented Eve-GAN, a novel domain translation task for inferring eavesdropped signals, enabling end-to-end training of PGM, utilizing few-shot learning techniques to reduce data collection overhead.
- Achieved a protection rate of over 97% against audio classifiers, hindering eavesdropped reconstruction.
- Paper published in the [IEEE Symposium on Security and Privacy 2025](#).
- **Illinois Mathematics Lab (formerly Illinois Geometry Lab)**, Student Scholar *Jan 2024 - present*
 - Advised by Professor AJ Hildebrand.
 - Researched the data science perspective of sports matches, analyzing the predictability of outcomes by comparing prediction accuracies across polls, the betting market, Elo ratings, and mathematical models like Bradley-Terry.
 - Presented findings at Rose-Hulman Undergraduate Math Conference 2024, UIUC Undergraduate Research Symposium 2024, and Joint Mathematics Meeting 2025.
 - Paper in submission to the [Mathematics and Sports \(MAS\) Journal](#).
- **Carnegie Mellon University Human-Computer Interaction Institute**, Research Intern *May 2025 - Feb 2026*
 - Advised by Professor [Jeff Bigham](#).
 - Designed and evaluated an automated journalism pipeline using SOTA LLMs to replicate end-to-end editorial workflows, from transcript segmentation to headline generation and topic prioritization.
 - Orchestrated large-scale crowd-sourced study and demonstrated that LLM-based headline quality and topic prioritization can meet and exceed professional standards.
 - Paper in CHI 2026 [Human-centered Evaluation and Auditing of Language Models \(HEAL\) workshop](#).
- **National University of Singapore SEEDER Group**, Research Intern *May 2024 - Aug 2024*
 - Advised by Professor [Xuanyao Fong](#).
 - Developed machine learning pipeline to detect, classify, and analyze diseases in plant leaves using U-Net segmentation and CNN classification models, achieving 90% segmented classification rate.
 - Stacked deep learning models with traditional ML models and augmented training data with cGAN-generated synthetic images, achieving over 95% detection accuracy.

PRESENTATIONS

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| 2025 | Carnegie Mellon University Human-Computer Interaction Institute REU Symposium LLMs for Extracting Agenda Items from City Council Meetings and Generating Newsworthy Headlines |
| 2025 | Algebra, Geometry and Combinatorics Day RSK linear operators and the Vershik-Kerov-Logan-Shepp curve (joint work presented by Duy Phan) |
| 2025 | Joint Mathematics Meeting Pi Mu Epsilon Poster Session Predictability in Major Sports Leagues |
| 2024 | Illinois Mathematics Lab Research Symposium Predictability in College Sports |
| 2024 | University of Illinois Undergraduate Research Symposium Predictability in College Sports |
| 2024 | Rose-Hulman Undergraduate Math Conference Predictability in College Sports |

RELEVANT COURSEWORK

CS: Machine Learning, NLP, Deep Learning, Algorithms, Numerical Methods, Numerical Analysis
Math: Honors Real Analysis, Graph Theory, Combinatorics, Abstract Linear Algebra, Abstract Algebra I-II, Algebraic Combinatorics, Linear Programming
Stats: Stochastic Processes, Applied Bayesian Analysis, Survival Analysis

GRANTS AND AWARDS

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| 2025 | CMU HCII REU: NSF 2349558 |
| 2025 | ICLUE RTG: NSF 1937241 |